

Dimensionality Reduction with PCA — Complete Step Summary

A clear, notebook-by-notebook HTML guide for the Wine Quality PCA, RNA-Seq PCA + Logistic Regression, and Expert PCA vs t-SNE vs UMAP exercises.

3

Notebooks
completed

20,531

RNA-Seq gene
features

530

Components
for 95%
variance

0.9415

Best
trustworthiness:
t-SNE

1

Core Concepts at a Glance

Dimension

A dimension is one feature or column. In Wine Quality, **alcohol**, **pH**, and **density** are dimensions. In RNA-Seq, each gene is a dimension.

Direction

Direction means the way the data mainly moves. PCA searches for the strongest new direction where the data is most stretched.

Spread / Variance

Spread means how far apart the values are. Variance is the mathematical measurement of that spread.

Noise

Noise is messy variation that does not clearly support the main pattern. PCA can sometimes reduce small noisy directions.

Standardisation

Standardisation puts features on a fair scale. It prevents large-number features from dominating PCA unfairly.

Projection

Projection means giving every row a new address in PCA space, such as PC1 and PC2 instead of many original columns.

Loadings

Loadings show how strongly each original feature contributes to a principal component. Large absolute values matter most.

Scree Plot

A scree plot shows cumulative explained variance and helps decide how many components to keep.

Trustworthiness

Trustworthiness checks whether nearby points in the 2D plot were also nearby in the original high-dimensional data.

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Big Picture: What PCA Does

Simple explanation

PCA takes many original columns and creates fewer new columns called **principal components**.

These components are mixtures of the original features and are ordered by how much variance they capture.

PCA is like finding the best camera angle for the data: it keeps the biggest visible structure while reducing the number of dimensions.

Many features

genes / wine chemistry

Standardise

fair scale

Goal: fewer dimensions while keeping maximum usefulness

Notebook 1 — Beginner PCA: Wine Quality

Goal

Use PCA to reduce 11 wine chemistry features into 2D and 3D visualisations. The **quality** column is used only for colouring, not for calculating PCA.

1,599

wine samples

11

input chemistry features

45.68%

variance in 2D PCA

59.78%

variance in 3D PCA

Steps completed

STEP 1

Loaded the Wine Quality dataset directly from UCI.

STEP 2

Separated X as wine chemistry features and y as quality labels.

STEP 3

Standardised the input features using `StandardScaler`.

STEP 4

Simple interpretation

The PCA plots give a simplified view of wine chemistry. Quality labels overlap strongly, especially common qualities such as 5 and 6. This means the first 2 or 3 components do not clearly separate wine quality.

Important: PCA did not use quality to create PC1, PC2, or PC3. Quality was added later only as a colour label.

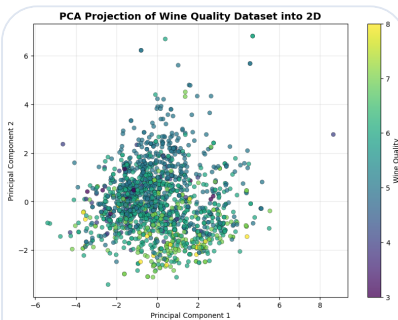
Applied PCA with 2 components and created a 2D scatter plot.

● **STEP 5**

Applied PCA with 3 components and created a 3D scatter plot.

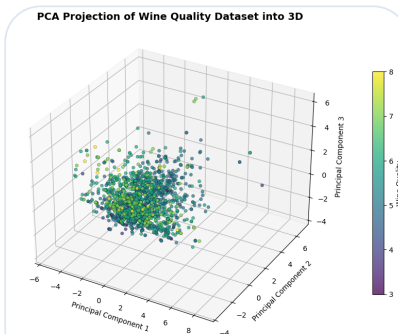
● **STEP 6**

Compared total explained variance for 2D and 3D PCA.



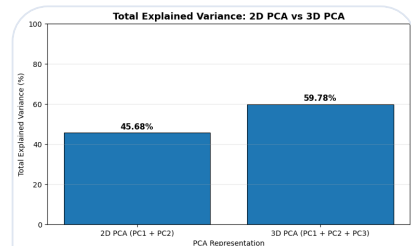
Wine PCA 2D Plot

PC1 and PC2 capture 45.68% of total variance. Quality classes overlap strongly.



Wine PCA 3D Plot

Adding PC3 increases total explained variance to 59.78%.



2D vs 3D Variance

The 3D PCA view preserves more structure, but still remains simplified.

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Notebook 2 — Intermediate PCA: RNA-Seq Scree Plot + Logistic Regression

Goal

Use PCA to reduce a high-dimensional gene expression dataset, decide the number of components needed for 95% variance, and compare Logistic Regression performance before and after PCA.

801

samples

20,531

gene features

530components for 95%
variance**100%**

PCA-reduced model accuracy

Steps completed

DATA

Loaded `data.csv` and `labels.csv` from the RNA-Seq dataset.

CLEAN

Removed the sample ID column and

Key results

MODEL	INPUTS	ACCURACY	TRAINING TIME
Logistic Regression — Full Features	20,531	98.14%	0.56s
Logistic Regression — PCA Reduced	530	100.00%	0.03s

PCA strongly reduced dimensionality while maintaining excellent classification

kept gene columns as X.



SCALE

Standardised all gene expression features.



PCA

Applied PCA with all possible components and created a cumulative variance scree plot.



95%

Found that 530 components explain approximately 95.01% of variance.



MODEL

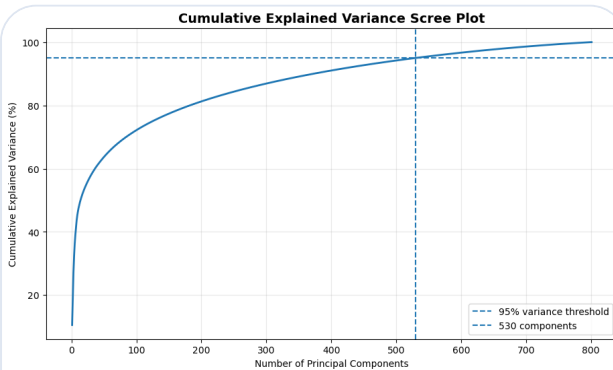
Compared Logistic Regression using full features vs

performance. PCA preparation time must still be considered separately.

PCA-reduced features.

LOADINGS

Analysed PC1 loadings to identify genes contributing most to the first component.



RNA-Seq Cumulative Variance Scree Plot

530 components are needed to preserve about 95% of the total variance.

Top PC1 genes

The strongest contributors to PC1 included genes such as gene_19862, gene_17360, gene_13489, gene_15158, and gene_7031.

The absolute loading values were close to each other, suggesting PC1 is influenced by a group of genes rather than one single dominant gene.

Loadings are not the same as supervised feature importance. They show contribution to PCA components, not direct predictive importance.

Notebook 3 — Expert PCA vs t-SNE vs UMAP

Goal

Compare linear and non-linear dimensionality reduction methods on the RNA-Seq dataset using visual separation, runtime, and trustworthiness.

0.40s

PCA runtime

4.01s

t-SNE runtime

19.58s

UMAP runtime

0.9415

t-SNE trustworthiness

Method notes

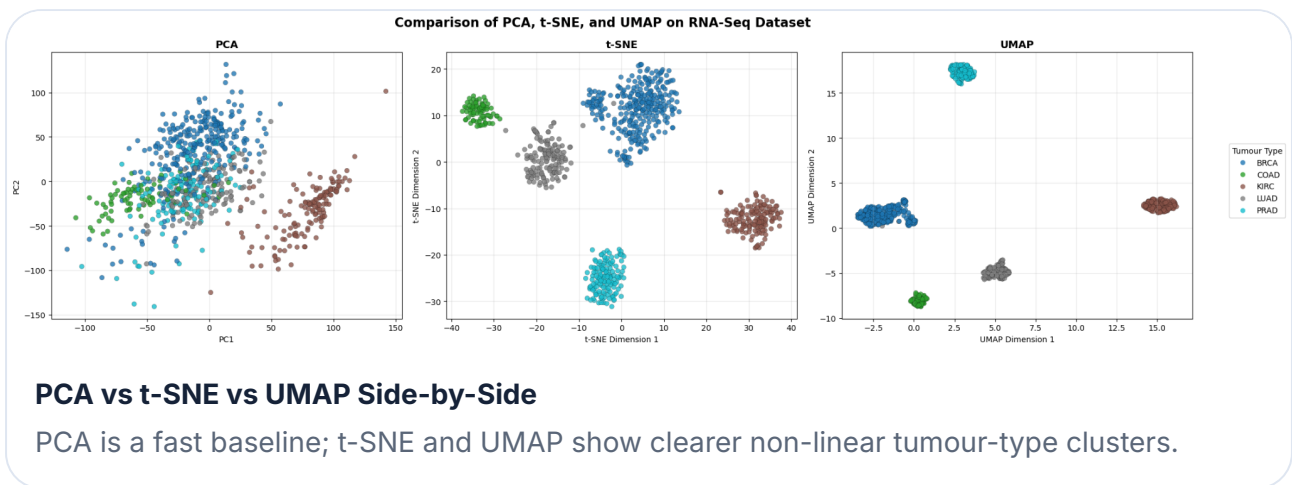
METHOD	SIMPLE EXPLANATION	OBSERVED RESULT
PCA	Best straight 2D view based on variance.	Fastest, but more class overlap.
t-SNE	Non-linear method that preserves local neighbours.	Strong cluster separation and best trustworthiness.
UMAP	Non-linear neighbour-	Compact visual clusters; slower in this run.

Trustworthiness scores

METHOD	TRUSTWORTHINESS
PCA	0.7863
t-SNE	0.9415
UMAP	0.9183

Higher trustworthiness is more reliable for nearby points.

METHOD	SIMPLE EXPLANATION	OBSERVED RESULT
	preserving method.	
PCA(50)+UMAP	Compress first, then run UMAP on 50 PCA components.	Final notebook contains the workflow cells for this comparison.



Runtime note: Exact runtimes can vary across environments. In this run, UMAP was slower than t-SNE, even though UMAP is often faster in other settings. This is normal and depends on package version, CPU, memory, and initialisation overhead.

Final Workflow Checklist

For PCA visualisation

1. Choose numeric input features.
2. Keep labels separate for colouring.
3. Standardise features.
4. Apply PCA with 2 or 3 components.
5. Plot PC scores and colour by label.
6. Report explained variance.

For PCA modelling

1. Split train/test before fitting preprocessing.
2. Fit scaler on training data only.
3. Fit PCA on training data only.
4. Transform both train and test.
5. Train model on PCA components.
6. Compare accuracy and training time.

For expert comparison

1. Run PCA, t-SNE, and UMAP.
2. Measure runtime for each method.
3. Plot embeddings with consistent colours.
4. Calculate trustworthiness.
5. Test PCA preprocessing before UMAP.
6. Write balanced interpretation.

Professional interpretation template

Good wording: "The PCA plot provides a simplified view of the dataset. Some class structure is visible, but overlap remains because PCA is unsupervised and only the first few components are shown."

Avoid: "PCA proves the classes are separated." PCA does not use labels and does not prove class separation by itself.

Created as a project-ready HTML summary for Day 5 PCA notebooks. Values are based on the notebook outputs from this session and may vary slightly if cells are rerun on another machine.